The Admissible Radius Is Tight: a Correction and a Measured Boundary in Online Conformal Prediction

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Abstract

Conformal PID states long-run coverage for an arbitrary bounded scorecaster, and a published corollary retains its rate under any predictable perturbation of the deployed threshold inside an admissible radius. The opposite reading is in print too, under overlapping authorship, and acted on. We make two claims. The record holds both halves and we join them, conceding placement and derivation by name. And the corollary's condition is tight: a legal L_1 dead band on the *completed* threshold leaves that set at $\tau^\star = \sup r_t + \sup \hat{q}_t - b/2$, its own radius at the null scorecaster, past which realised miscoverage is 1.000000 against $\alpha = 0.1$ while saturation fires every round. Leaving it costs coverage, not only the rate: partial adjustment there overruns Proposition 2's bound $42\times$ at $T = 10^6$ and covers, 0.100035 at 2×10^5. That edge belongs to the saturator–scorecaster pair: under $\hat{q} \equiv +b/2$, equally legal, the failing band covers; under $\hat{q} \equiv -b/2$ both readouts return 1.000000.

1 Introduction

An online conformal method emits a threshold before each observation and moves it afterwards. Forecast stability names that movement, scores it and prices it [Tunc et al., 2013, Godahewa et al., 2025, Van Belle et al., 2023, Pritularga and Kourentzes, 2024, Genov et al., 2026, Van Belle et al., 2026]. Varying a forecast's temporal structure at fixed level was invented earlier, in forecast verification, where the predictive marginal is matched on real producers [Gneiting et al., 2007, Pinson and Girard, 2012, Worsnop et al., 2018]; that coverage and mean length under-describe a deployed interval is established too [Min et al., 2026, Vaze, 2026]. None of it is claimed here.

First: the record carries a claim and its refutation, and this paper joins them. Conformal PID [Angelopoulos et al., 2023] adds a *scorecaster* $\hat{q}_{t+1}$ to a saturating error integrator and asks of it only that it be predictable and lie in $[-b/2, b/2]$ — "any function of the past" (Theorem 1) — so a readout placed in that slot is already quantified over. Also in print, of the same method: "[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical coverage guarantee" [Li and Rodríguez, 2024] — stated and acted on in that paper's baselines paragraph, refereed and verbatim on printed page 18443. The refutation is in print too, twice and with proof: Li et al. [2025, Corollary 2] add an arbitrary predictable d_t with $|d_t| \le \mu_t(b/2 - |\hat{q}_t|)$ to the deployed threshold, observe that the result "has exactly the CPID error-integration form", and conclude $|E_T| \le c\,h(T) + 1 = o(T)$; the same generic-slot argument carries AcMCP's long-run coverage [Wang and Hyndman, 2024]. The first and last authors of Li et al. [2025] are the authors of Li and Rodríguez [2024] and cite it: the halves already share a bibliography, this is ordinary self-correction, and the joining is what this paper adds. Four surfaces, queried on 20 August 2026 — arXiv full text and abstracts, OpenReview, and the six works Semantic Scholar records as citing it — return zero printing both. Hu et al. [2026] still calls scorecaster choice "arbitrary" and lacking "principled guidance", while asserting valid coverage: model selection, not validity. **We claim neither the placement nor the derivation.**

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Second: that admissible set has an edge, and the condition there is tight. Corollary 2 is a *retention* result, so what it leaves open is outside that set. Section 3 exhibits a legal perturbation that leaves it — an L_1 dead band on the *completed* threshold, the family characterised here — and locates the edge at $\tau^\star = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, the corollary's own radius at the null scorecaster. Past it the price is coverage and not only the rate, so the published sufficient condition is necessary there. Those two claims are the paper; Section 4's four vocabularies for the same movement are description, not a third claim.

2 Setup

The producer. A forecaster fit on data before t induces a score s_t, a threshold q_t is emitted before y_t is revealed, the deployed set is $C_t = \{y : s_t(x_t, y) \le q_t\}$ and $\mathrm{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$; the target is long-run coverage, $T^{-1} \sum_{t \le T} \mathrm{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID [Angelopoulos et al., 2023] manipulates the threshold on the score scale,

$$q_{t+1} \; = \; \hat{q}_{t+1} \; + \; r_t\Big(\textstyle\sum_{i \le t} (\mathrm{err}_i - \alpha) \Big), \tag{1}$$

their iteration (5), in which r_t saturates: condition (4) requires $|r_t(x)| \ge b$ with the sign of x once $|x| \ge c\,h(t)$, for h nonnegative, nondecreasing and sublinear. Writing $E_t = \sum_{i \le t} (\mathrm{err}_i - \alpha)$, Proposition 2 gives $|E_T| \le c\,h(T) + 1$; boundedness is read on realised scores.

One scalar, two readouts, two places to put it. A movement penalty on a sequentially updated scalar has two standard readouts, both prior work: a quadratic cost gives linear partial adjustment towards the target, $\tilde{q}_{t+1} = (1 - \lambda)\hat{q}^{\mathrm{raw}}_{t+1} + \lambda \tilde{q}_t$ [Gârleanu and Pedersen, 2013], and a proportional cost a no-trade band, moving only when the target is more than τ away and then pulling back by exactly τ [Constantinides, 1986, Davis and Norman, 1990]. The first is published as post-processing [Godahewa et al., 2025] and applied to a deployed conformal threshold [Binny and Dixit, 2025, Eq. 13]. It has two positions in (1): **(B)** in the $\hat{q}_{t+1}$ slot, where both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster's range, inside Corollary 2's admissible set; and **(A)** downstream of the completed q_{t+1}, outside it. Placement A is measured here. *The dead band (L1) is not a stylistic alternative to the partial-adjustment (L2) form*; it is the specific perturbation family whose failure §3 characterises. Neither is safe by construction: L2 forfeits Proposition 2's rate while covering, L1 past $\tau^\star$ loses coverage, and at the legal $\hat{q}_t \equiv -b/2$ both lose it. What is tight is the radius, not a form.

The quantity, and the names it already has. We report the deployed threshold's *travel*, $\mathcal{T}_H = \sum_{t=2}^{H} |q_t - q_{t-1}|$, after actuator travel in process control. The arithmetic is not new: $\sum |\Delta q|$ over a predictive distribution is the Multi-Quantile Change [Zanotti, 2026, v3, §3.4.1]; verification calls it *jumpiness*, a *convergence index* and *(in)consistency* [Zsótér et al., 2009, Ehret, 2010, Pappenberger et al., 2011]; applied control, IACER [Sarhadi, 2025]; dynamic-regret online learning, a path length. Table 2's caption names ten across four vocabularies: every natural name is taken, which is why a fresh one is used here.

Harness. $\alpha = 0.1$, $b = 2$, $c = 1$, $h(t) = \log(t + 2)$, and the clipped saturator $r_t(x) = b\,\mathrm{clip}(x/(c\,h(t)), -1, 1)$, which meets condition (4) with equality. The scorecaster is null and legal, reducing (1) to the pure error integration Proposition 2 is stated about. Neither free choice is neutral: condition (4) is a lower bound this saturator meets exactly, and Theorem 1 permits scorecasters $b/2$ either side of the null one. The adversary is specified rather than tie-broken: it plays $s_t = \mathrm{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$, $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$ — the $\varepsilon \downarrow 0$ reading of "sets the score at the threshold", firing the strict indicator. Legality is Theorem 1's $[-b/2, b/2]$ on both sides, so $b/2 + b/2 = b$ is exactly tight.

3 The edge of the admissible set, measured

The retention result is published, and this section tests it from outside. Li et al. [2025, Corollary 2] prove that a predictable d_t with $|d_t| \le \mu_t(b/2 - |\hat{q}_t|)$ on the deployed threshold keeps (1) in error-integration form and *retains* $|E_T| \le c\,h(T) + 1$; the admissible radius is theirs. A downstream partial adjustment of weight w leaves it. Its excursion at $T = 10^4$ is

Table 1: Placement A: a one-scalar readout on the *completed* threshold; one deterministic adversary, one pass to $T = 10^6$. *All eleven arms run are listed*, so no covering arm is withheld. Proposition 2's bound is 13.2061 at $T = 2{\times}10^5$ and 14.8155 at $T = 10^6$; "ratio" is $\max_t |E_t|$ over it, "Sat." the rounds on which (4) delivers its full $\pm b$.

Readout on the completed threshold	miscov. $2{\times}10^5$	$\max_t |E_t|$ 10^6	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	63.00	4.25	0.0007	1,024
partial adj. $w = 0.999$	0.100035	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	53.00	3.58	0.3181	16
running mean	0.099940	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	**1.000000**	900,000	60,747	1.0000	0.5

$6.30/63.00/623.70$ for $w = 0.9/0.99/0.999$, and for the last two it is *flat in* T while the bound grows like $\log T$, so the ratio falls with the horizon, 61.08 to 42.10; the measured law across the arms is $\max_t |E_t| \approx \max\{0.5\, h(T) + 0.9,\ 0.63/(1-w)\}$, and five of the eleven arms exceed the bound at $T = 10^6$. The forfeit is therefore a property of the smoothing *strength* and not of Placement A as such, and it buys something: the $w = 0.999$ arm cuts deployed travel from 28,311 to 202. The sign is the opposite of the intuitive one. A readout on the completed q_{t+1} touches neither r_t nor the integrator's argument, so it cannot put condition (4) at risk; it delays the correction the integrator has already computed, so the accumulator excurses *further*. Smoothing the output does not damp E_t, it feeds it. The 623.70 also turns on the tie rule: it is the figure for an adversary playing the threshold *exactly* with ties scored as misses, and 70.80 under the strict indicator this paper uses elsewhere.

The L_1 dead band leaves the radius, and the edge is measurable. A dead band pulls the deployed threshold back by exactly τ when it releases, so "inside Corollary 2" reads $\tau \le \mu_t(b/2 - |\hat{q}_t|)$. Under saturation the deployed value is pinned τ below the integrator's ceiling: its realised supremum is $b - \tau$ to the printed digit at every $\tau \ge 0.9$. The adversary's best legal play is $b/2$, so coverage is lost exactly when that supremum drops below it, at $\tau^\star = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$. Every dead-band configuration in the run record agrees, 19 of 19: three widths under each of the null scorecaster, $\hat{q} \equiv +b/2$, $\hat{q} \equiv -b/2$, a saturator at level $4b$ and Angelopoulos et al.'s tangent integrator, plus four wider bands at the null scorecaster. Three widths per setting bracket each edge rather than locate it; a booked bisection in τ would locate it, and none is booked here. **Here** $\sup r_t = b$ **and** $\hat{q} \equiv 0$**, so** $\tau^\star = b/2$ — one point of the admissible class, and both terms move it. At $\hat{q} \equiv +b/2$, legal, the same $\tau = 1.5$ band covers at $T = 10^6$ (0.100010, $\max_t |E_t| = 11.20$, inside the bound); at $\hat{q} \equiv -b/2$, equally legal, $\tau^\star = 0$ and $\tau = 0.5$ returns 1.000000, as does partial adjustment at $w = 0.999$. Raising the saturator's level to $4b$, admissible since (4) is a lower bound, raises $\sup_x r_t(x)$ and $\tau^\star$ with it and cuts 623.70 to 120.60; under the unbounded tangent integrator Angelopoulos et al. [2023] report using in all their experiments, $\sup_x r_t(x)$ is unbounded and so is $\tau^\star$, the three widths run there all cover, $\tau = 1.5$ at 0.100001, and 623.70 falls to 20.10. **The method's own integrator therefore sits inside the set at every width run here.** Outside it the transition is a step: $\tau = 1.000$ covers at $T = 10^5$ while $\tau = 1.001$ returns 1.000000, and past $\tau^\star$ the threshold sits permanently below the reachable score range and $|E_t| = (1-\alpha)t$ grows linearly, 60,747 times the bound at $T = 10^6$; $\tau = 2.0$, 2.5, 3.0 and 5.0 fail too, so the failing set is $(\tau^\star, \infty)$, open at the left — the endpoint covers. **And the failing arms have** $\mathrm{Sat.} = 1.0000$: condition (4) delivers its full $\pm b$ on every round, and what fails is Proposition 2's other hypothesis, that the threshold closing the loop *is* the integrator's output.

What that measures is that the published condition is tight. At $\mu_t = 1$, $\hat{q}_t \equiv 0$ the measured $\tau^\star$ is exactly Corollary 2's radius, and crossing it costs coverage rather than the rate. **The claim is that necessity: a tightness result about someone else's sufficient condition**, smaller and more precise than a forfeit of the rate by post-processing. Elsewhere the condition stays sufficient only — at $\hat{q}_t \equiv +b/2$ Corollary 2 permits radius 0 while every width run there covers — which is what makes "tight" a located claim.

Table 2: Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness*, *convergence index*, *(in)consistency*; and *switching cost*. The row gives what each contributes.

Online conformal	**Forecast stability**	**Verification**	**Switching-cost OL**
a distribution-free long-run-coverage guarantee for *any* bounded scorecaster, with a rate [Angelopoulos et al., 2023, Thm 1, Prop. 2], and the admissible set measured here to be tight	the scored movement functional, the readout map as post-processing [Godahewa et al., 2025, Eq. 3], and a priced decision [Genov et al., 2026, §4.2], [Van Belle et al., 2026]	the matched-level design on real producers from 2012, with a control *stronger* than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]	worst-case theory for the penalised problem: Thm 2, and a Thm 7 trade-off in which sublinear regret is bought by growing θ with T, which grows the ratio [Andrew et al., 2013]

The constructive reading, and it is Li et al.'s rather than ours. Everything above is about leaving a sufficient condition, so the design consequences are readings of that condition and not results of this paper. Two follow. **Placement B is covered already**: the same readout applied inside the $\hat{q}_{t+1}$ slot keeps $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster's range, hence inside the admissible radius, so Theorem 1 carries it with no further argument — which is why it is not measured here. **And at Placement A the operative quantity is $\tau^\star$, which is partly a design variable.** Condition (4) bounds $\sup_x r_t(x)$ only from below, so raising the saturator's level widens the admissible band at no cost to the theorem: that is the whole content of the $4b$ and tangent rows above. What the measurement adds is that the band has an edge, that the edge is computable before deployment from $\sup_x r_t(x)$ and $\sup_t \hat{q}_t$, and that crossing it is not a graceful degradation — realised miscoverage goes to 1.000000.

4 Where this sits

Four vocabularies name one quantity many ways, and the object that settles its validity — the admissible set around the deployed threshold — sits in the first of them.

What is conceded, by name. The placement is prior work: Angelopoulos et al. [2023] state coverage for any admissible integrator and any scorecaster and run a smoothing-family model in that slot themselves; Dupuy et al. [2026] give the generic argument in their Appendix A and Duerst et al. [2024] adopt that scorecaster. The derivation is prior work too, and that is the concession that matters most: Li et al.'s Corollary 2 proves the retention statement in Conformal PID's own notation [Li et al., 2025], with Wang and Hyndman [2024] an independent second instance. The smoother object, the metric arithmetic and both readout maps are conceded in Section 2, and Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a feedforward path, where free-until-the-bound-binds is the defining specification rather than a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model* update on the conformal score; their one non-standard assumption is that $\sum_{t\leq T} q_t \leq O(\alpha\, h(T))$ for a sublinear h, supported with a plot rather than derived. That accumulation is what Section 3 measures diverging.

5 Limitations

One construction, one adversary, one saturator, and $0.63/(1-w)$ is specific to its *level* rather than to its shape, and condition (4) bounds that level only from below. **The inherited guarantee is weak where it applies.** Under the tangent integrator $h(t) = t/\log t$, so the certified gap $(c\, h(T) + 1)/T$ decays as $O(1/\log T)$ rather than $O(1/T)$: what is forfeited was not a tight certificate. **The dead-band widths are absolute, and the failure is total only against the adversary.** τ is in score units with $b = 2$, so the failing $\tau = 1.5$ is $0.75\, b$; under i.i.d. scores that arm returns 0.249376 rather than 1.000000. What survives is the mechanism: a band wider than $\tau^\star$ caps the reachable threshold below the reachable score range. **The harness was rebuilt against numbers already read**, which carries less weight than a rerun done without sight of them; the free choices are registered in the code. **Placement B, the readout Corollary 2 already covers, is not measured here**: it is conceded rather than tested, and this paper offers no evidence about it beyond the published argument. **One seed, and the grid brackets rather than locates.** Every arm is a single pass at one seed; the three widths per setting bracket each $\tau^\star$ but no bisection is booked, so the edge is located to within a tested interval and not to a digit. **What would overturn the tightness claim** is a legal $(r_t, \hat{q}_t)$ pair at which a dead band of width $\tau \leq \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$ loses coverage, or one above it that keeps coverage; the run record contains neither, over the nineteen configurations it covers.

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